Practical 4a

Bus Topology:

Bus topology is a type of network topology in which all devices are connected to a single cable called a bus. This cable serves as a shared communication medium, allowing all devices on the network to receive the same signal simultaneously.

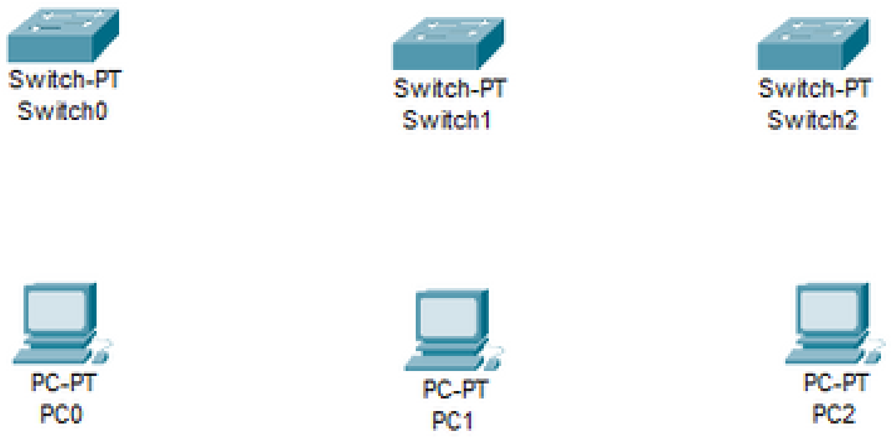
Ring Topology:

Ring topology is a type of network configuration where devices are connected in a circular manner, forming a closed loop. In this setup, each device is connected to exactly two other devices, creating a continuous pathway for data transmission.

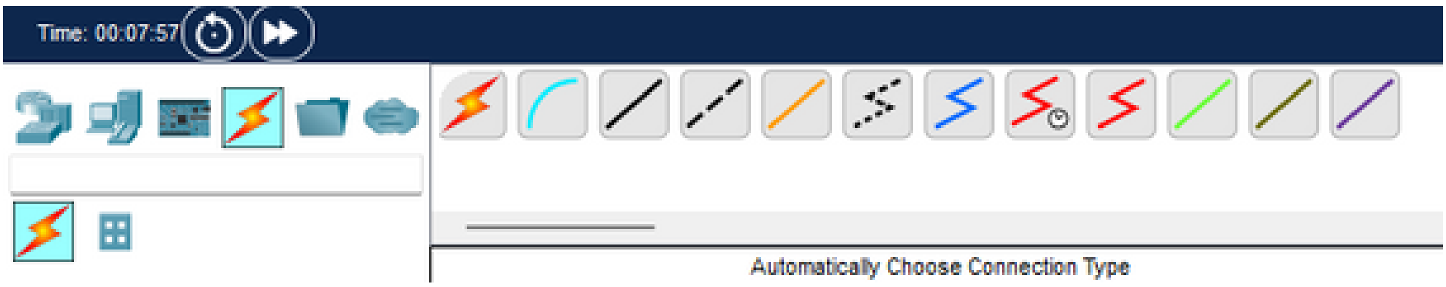
Steps of simulation of bus topology:

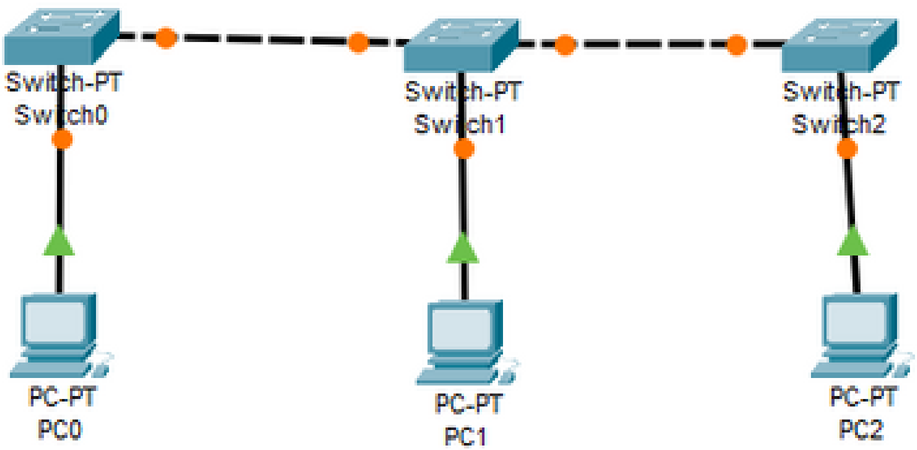
1. Add end Devices PCs (PC0 ,PC1 ,PC2) by clicking on end devices from the left panel.

2)Add a switch by clicking on network devices and drag and drop a switch into the workspace.



1. Connect the PCs to switch by clicking on connection and select on the copper straight-through cable.





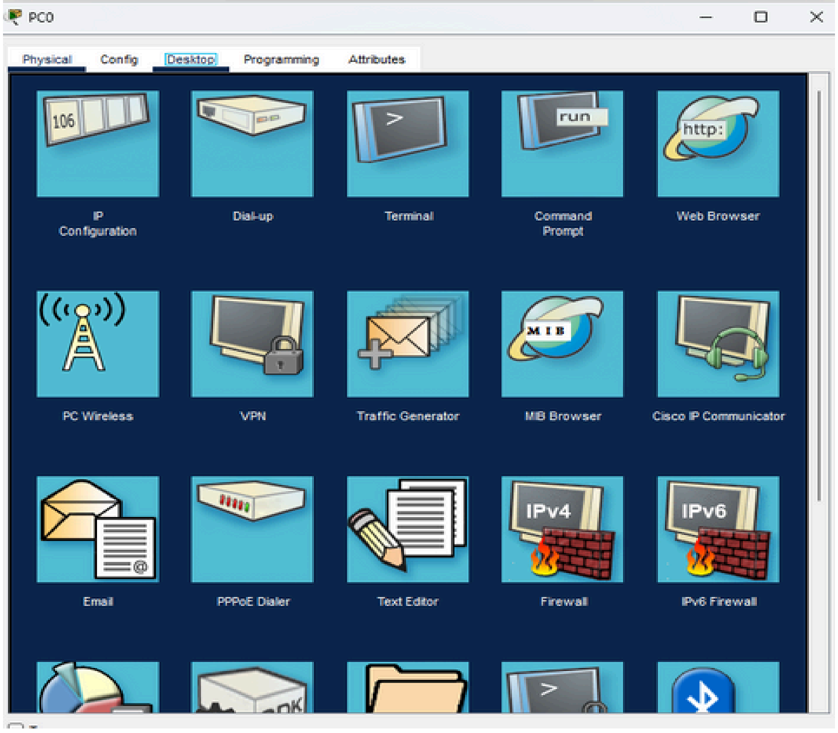
1. Assign the IP addresses tp PCs by clicking on a PC and go to desktop and go to IP configuration .

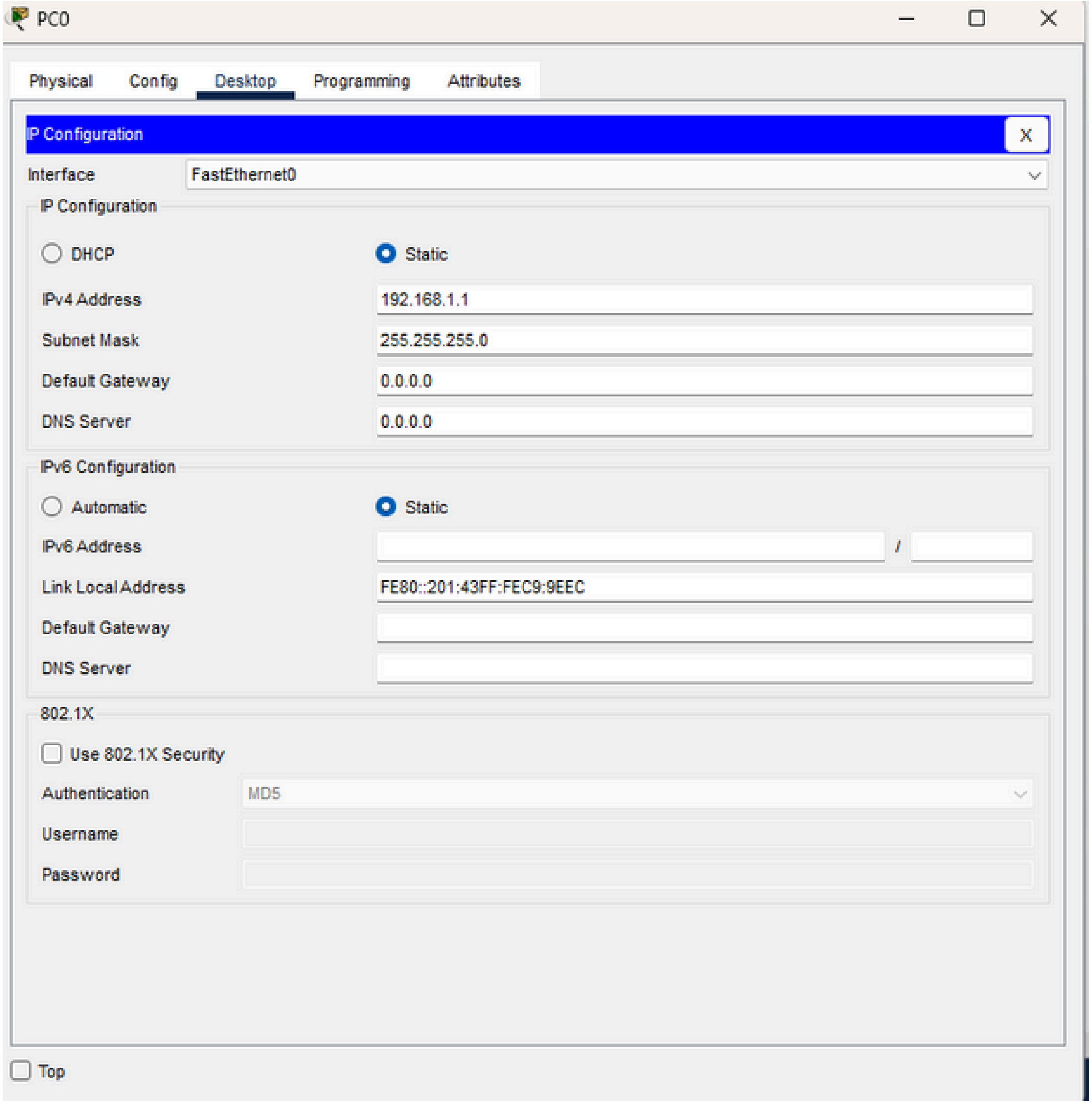
assign address in the same network

PC0: 192.168.1.1

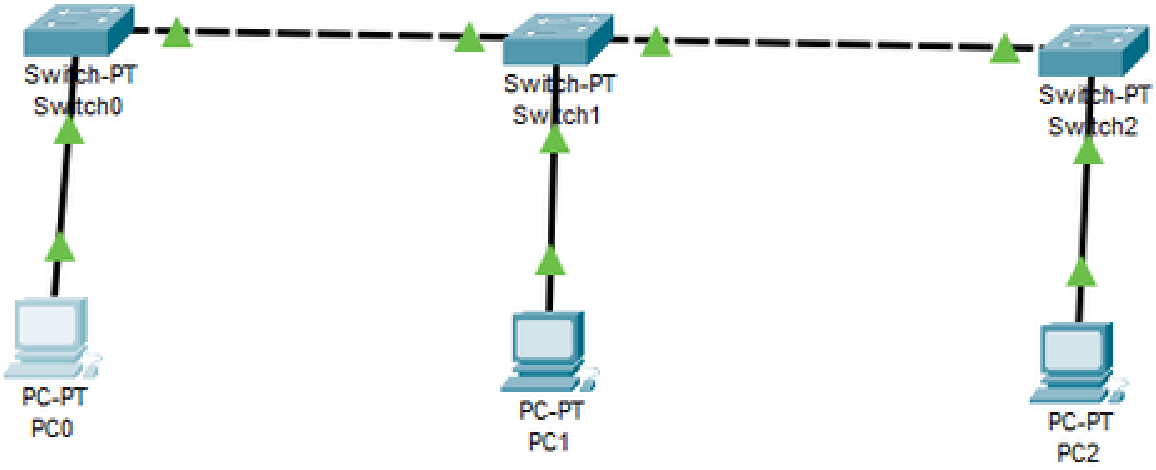
PC1: 192.168.1.2

PC2: 192.168.1.3

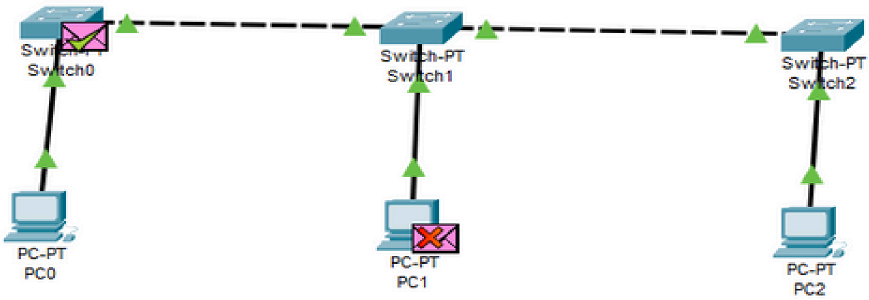

PC0
Physical
Config
Desktop
Programming
Attributes
106
IP Configuration
Dial-up
Terminal
run
Command Prompt
http:
Web Browser
PC Wireless
VPN
Traffic Generator
MIB
MIB Browser
Cisco IP Communicator
Email
PPPoE Dialer
Text Editor
IPv4
Firewall
IPv6
IPv6 Firewall

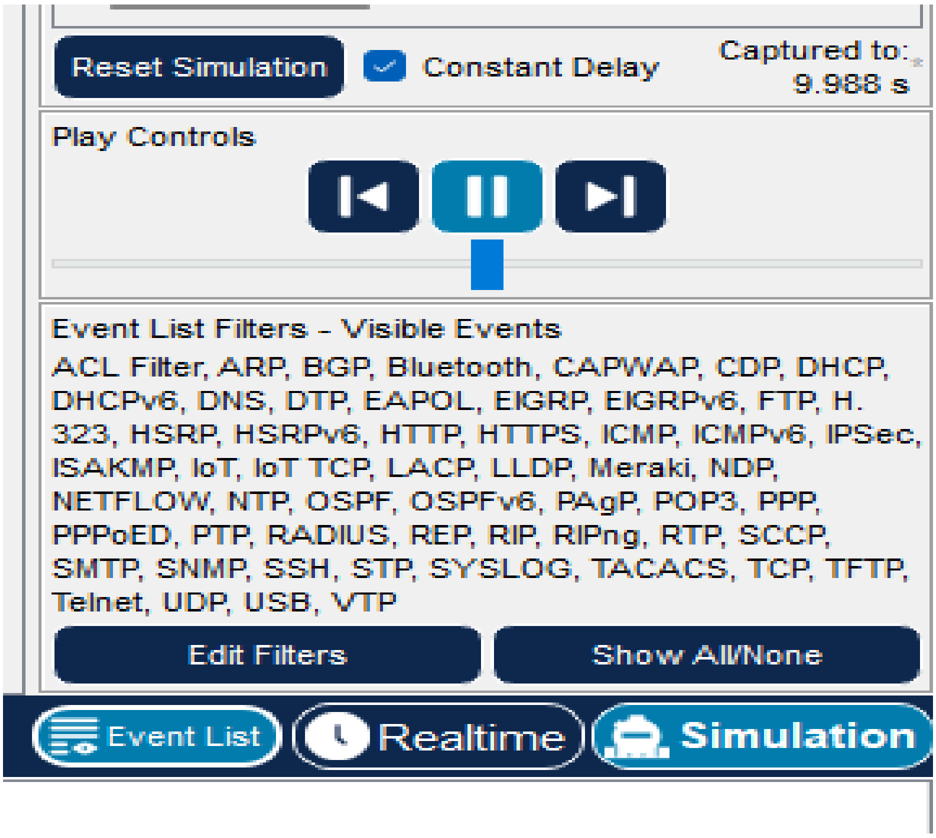


5)verify the connection by ensure the all links turns green.

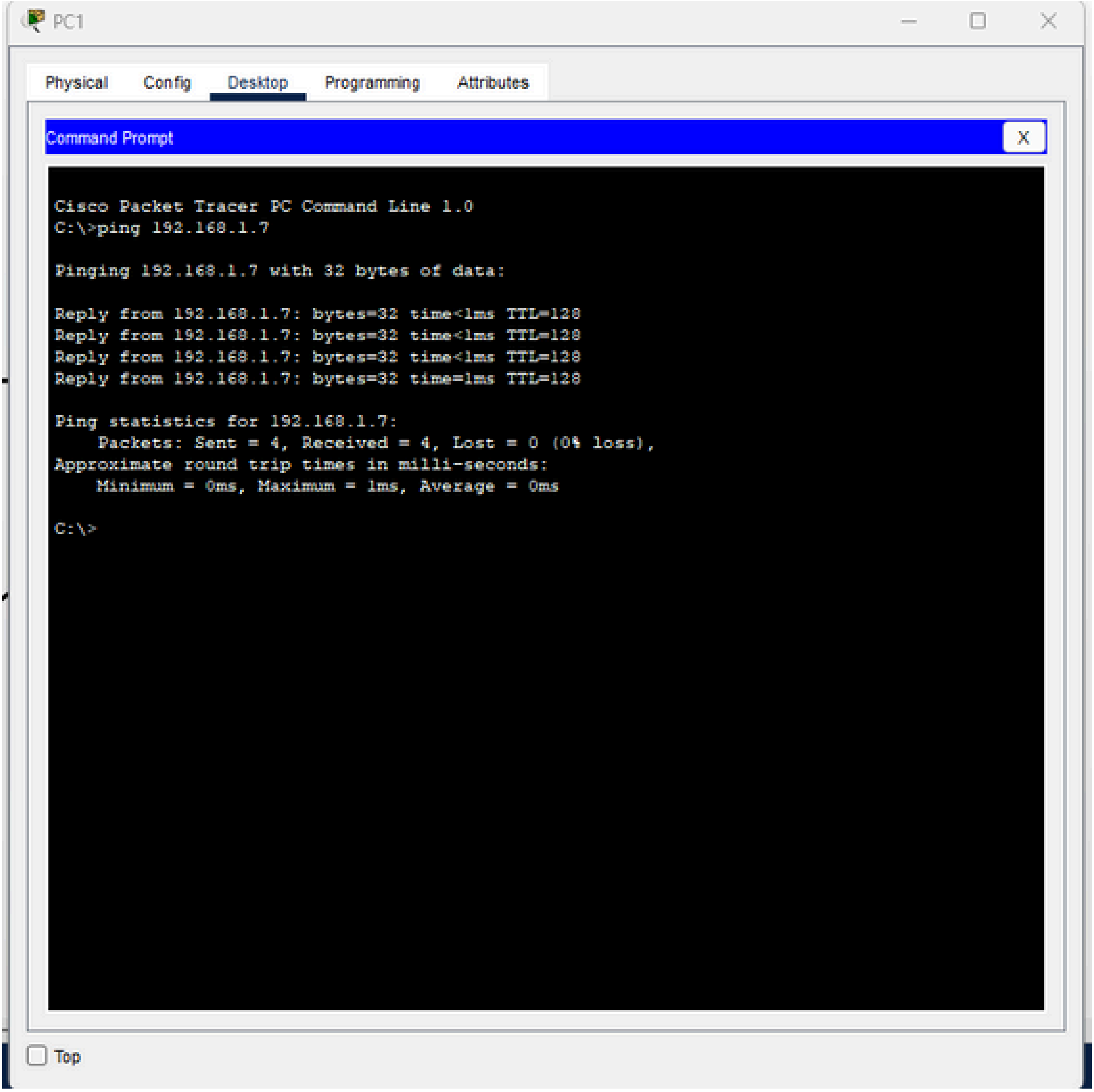


6)Test the network by sending the massage and click on simulation and click on run button to see the massage transfer.

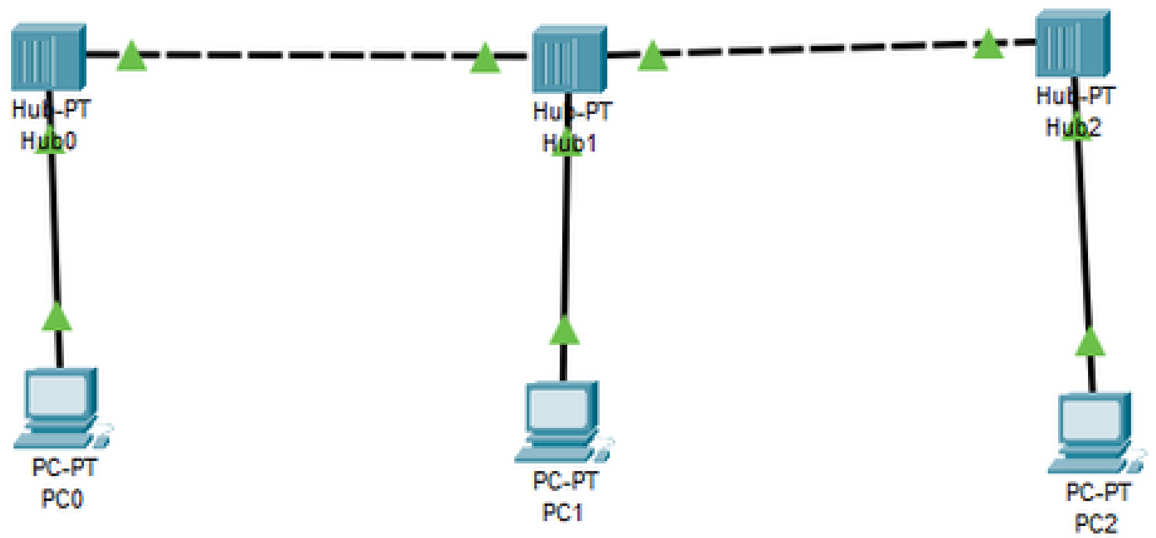


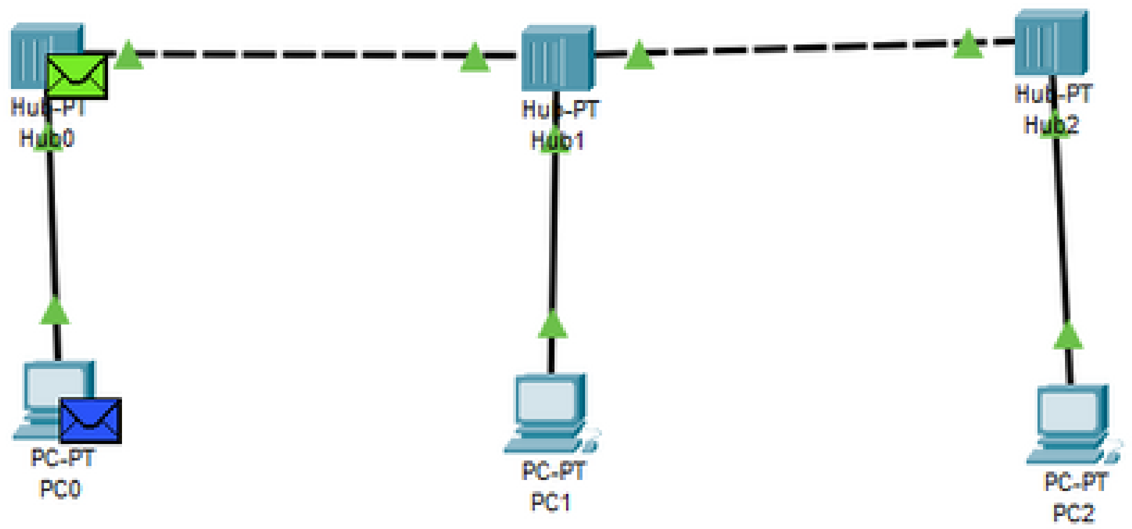


7) Also test the connection by clicking on PC0 go to the desktop and select the command prompt and type: ping 192.168.1.7 .If replies received the connection is successful.



8)Now replace the switch by hub.





Steps of simulation of ring topology:

1. Add end Devices PCs (PC0 ,PC1 ,PC2) by clicking on end devices from the left panel.

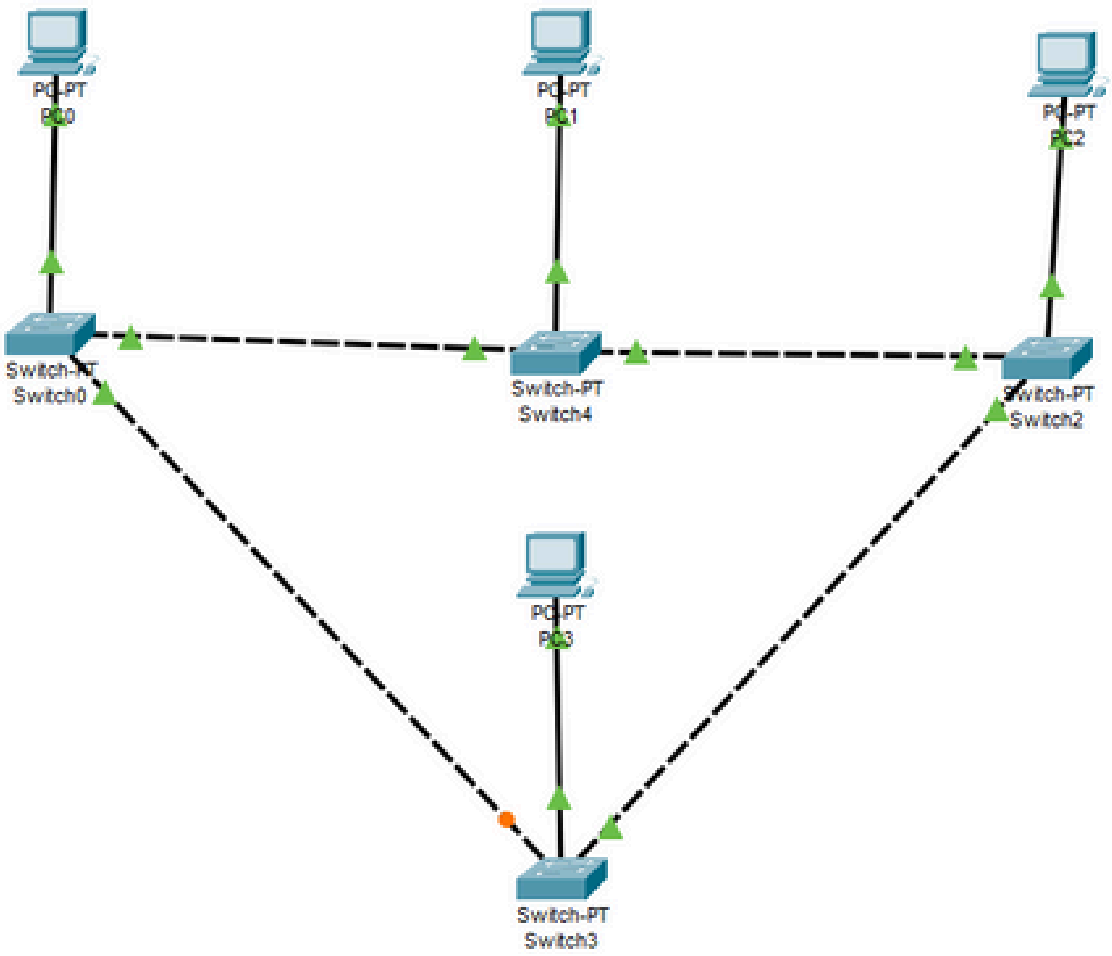
1. Connect PCs in a ring by clicking on connection.

Connect

switch 0 to switch 3

switch 3 to switch 2

switch 4 to switch 2



3) Assign the IP addresses tp PCs by clicking on a PC and go to desktop and go to IP configuration .

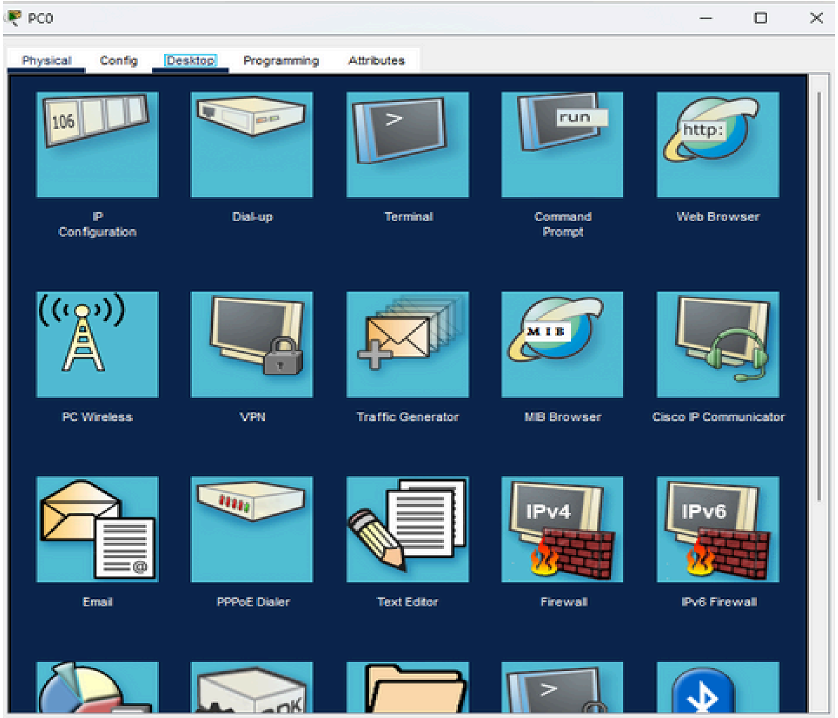
assign address in the same network

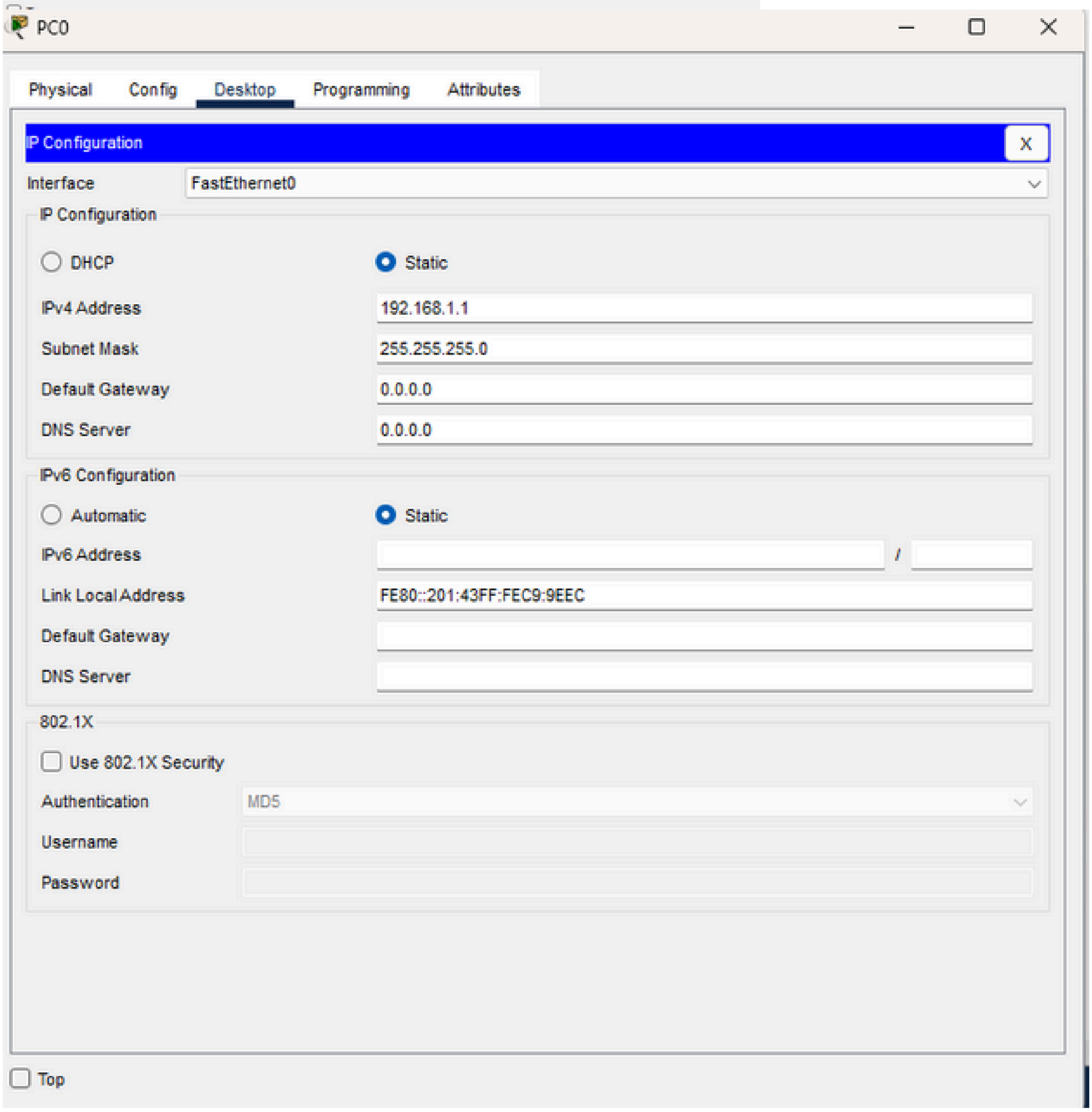
PC0: 192.168.1.1

PC1: 192.168.1.2

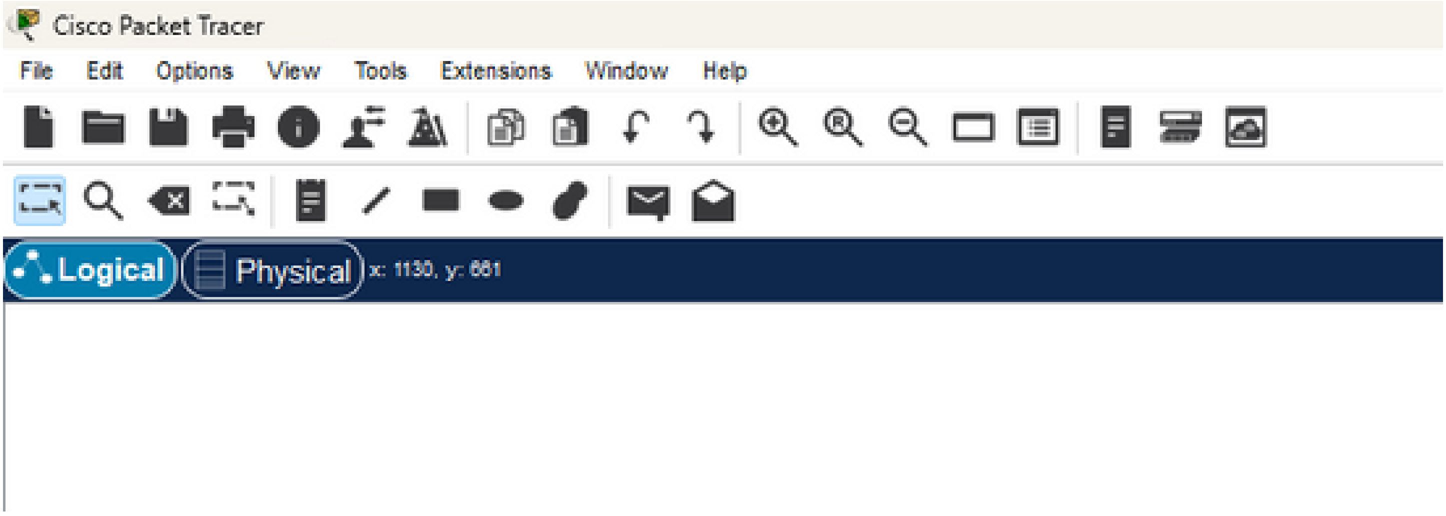
PC2: 192.168.1.3

PC3:192.168.1.4

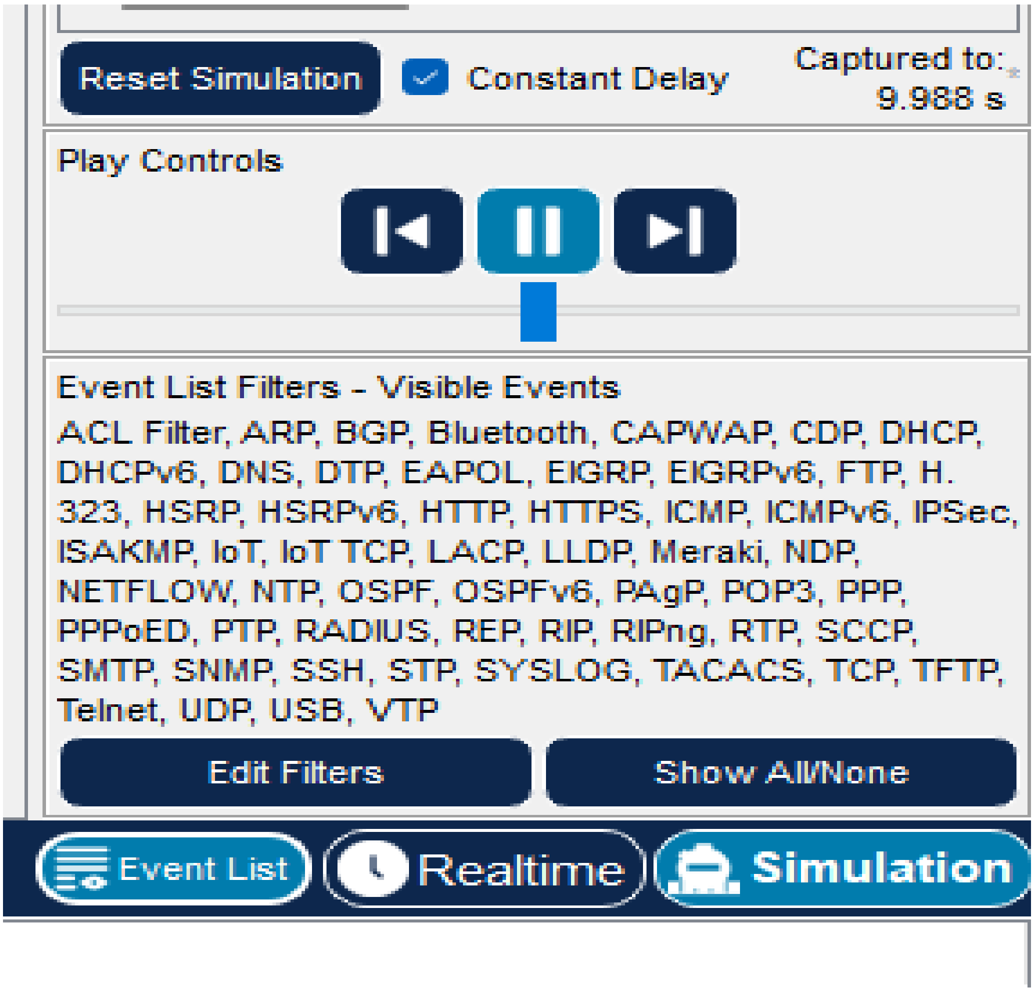


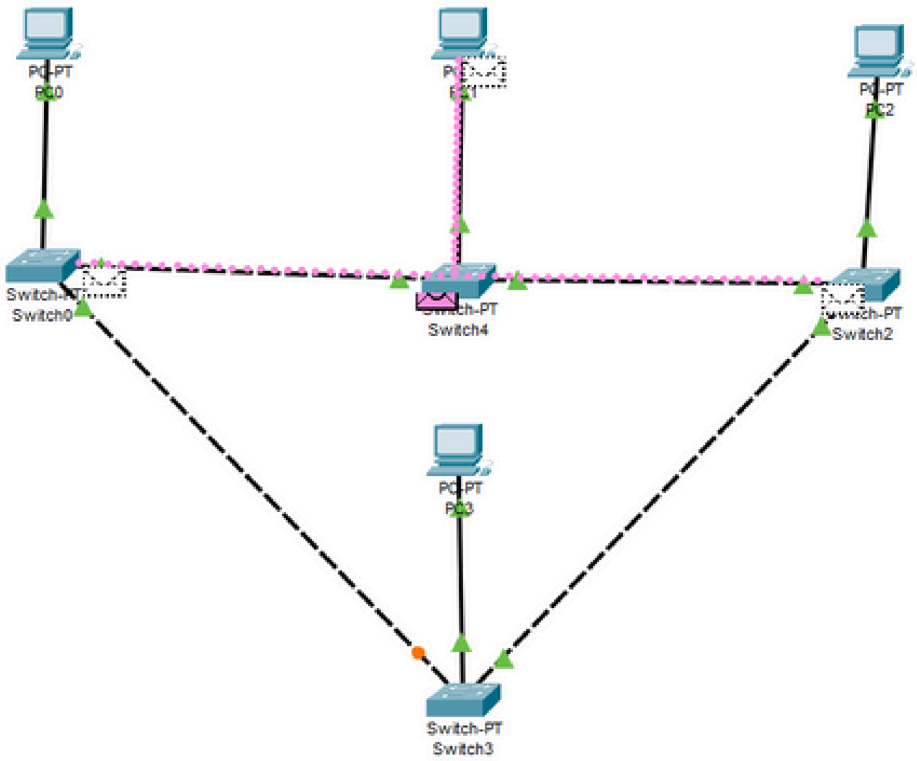


4)Now check the connection by sending the message from on PC to another PC.

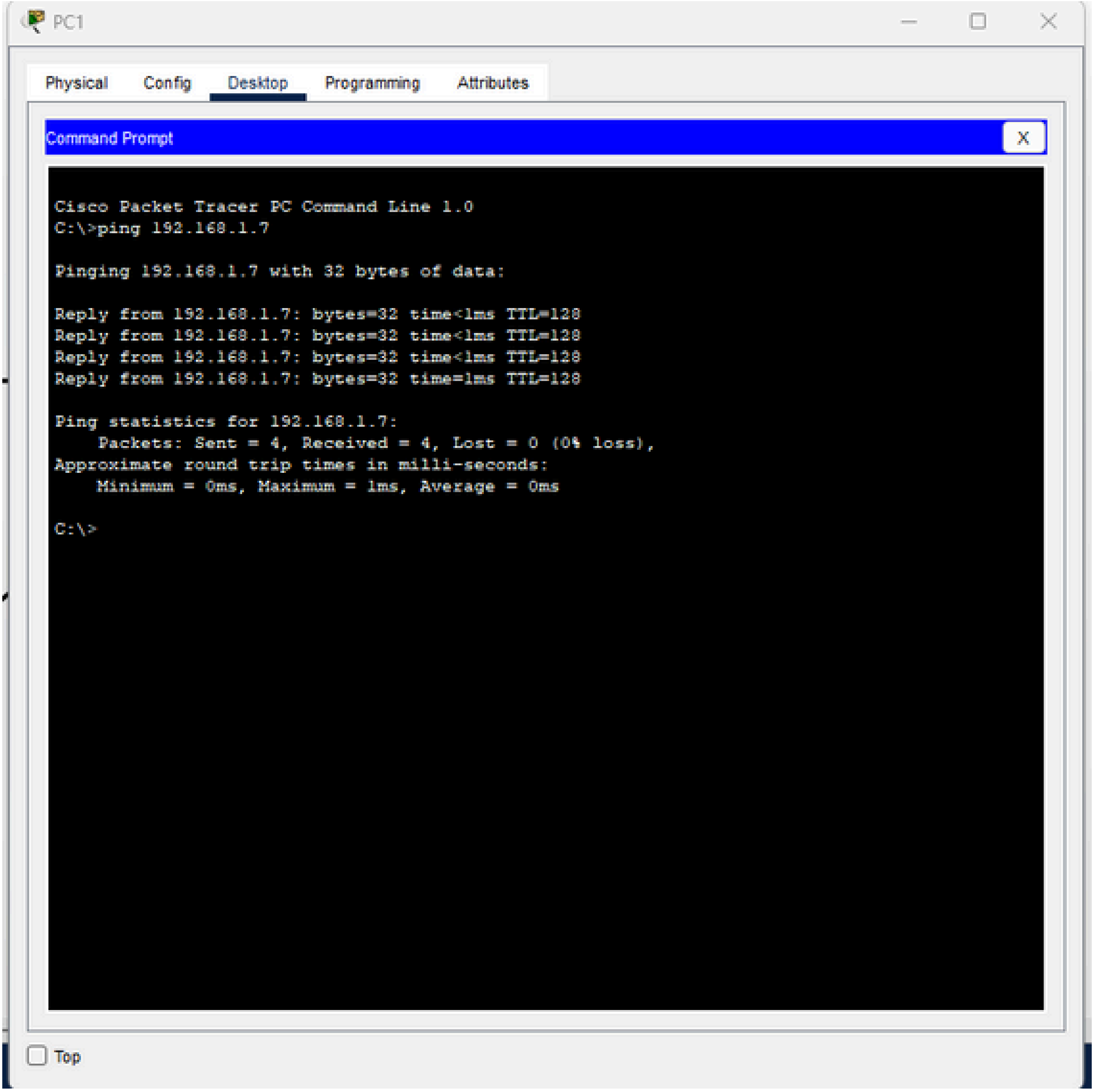


1. Click on simulation then click on run button to see the massage tranfer.




PC-PT
PC0
PC-PT
PC2
Switch-PT
Switch0
Switch-PT
Switch4
Switch-PT
Switch2
PC-PT
PC3
Switch-PT
Switch3

6) Also test the connection by clicking on PC1 go to the desktop and select the command prompt and type: ping 192.168.1.7 .If replies received the connection is successful.



Difference between bus and ring topology:

1. In bus topology, data is broadcast to all devices on the network simultaneously, while in ring topology, data is transmitted in a unidirectional loop.
2. In a Ring topology, each device is connected to two other devices in a circular fashion. Bus topology is a topology where each device is connected to a single cable which is known as the backbone.
3. Ring topology has a lower implementation cost than a bus topology.

The implementation cost of Bus topology is higher than Ring topology.